An Integer Topology Carrier Beyond Center Flux

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Abstract

We prove that the six mod-two center labels cannot retain integer four-dimensional flux pairing and define the minimal additional integer carrier in the abelian embedded calibration.

For integral fluxes $(n_{01}, n_{02}, n_{03}, n_{12}, n_{13}, n_{23})$, define

$$Q_{\mathrm{Pf}} = n_{01}n_{23} - n_{02}n_{13} + n_{03}n_{12}.$$

Theorem 1. *Reduction of the six fluxes modulo two does not determine Q_{Pf}. Indeed, the family*

$$(2k, 0, 0, 0, 0, 2k)$$

has fixed zero center label and charge $4k^2$. Hence integer charge is unbounded on one center-label fiber. Moreover Q_{Pf}^2 is the determinant of the associated antisymmetric flux matrix.

Proof. Direct substitution gives zero mod-two flux and charge $4k^2$. The determinant identity is the standard four-dimensional Pfaffian formula and also follows by expansion of the displayed antisymmetric matrix. □

The retained topology state must therefore include an integer lift in addition to the center label.

Remark 1. *This is an exact integral-flux and embedded-abelian calibration. It is not a construction of the full nonabelian lattice second Chern class, nor a sector-weight estimate.*